Computational Physics
Final Term Project

# NUMERICAL SOLUTIONS OF THE LORENZ SYSTEM HIGHLY DEPEND ON THE TIME STEP SIZE

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# 1 Mathematical Background

## 1.1 Malkus-Lorenz water wheel

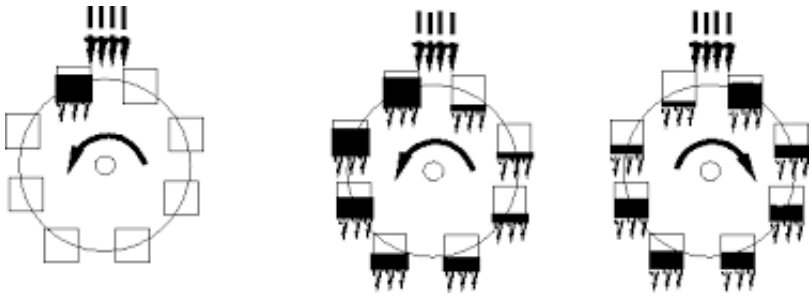

Figure 1: Discrete Water Wheel Model

The above figure is a discrete case of Malkus-Lorenz waterwheel. It has $N$ countable baskets, which have the same size with each other. In this case, water can not transport from one basket to another basket. ((a) in Fig. 2)

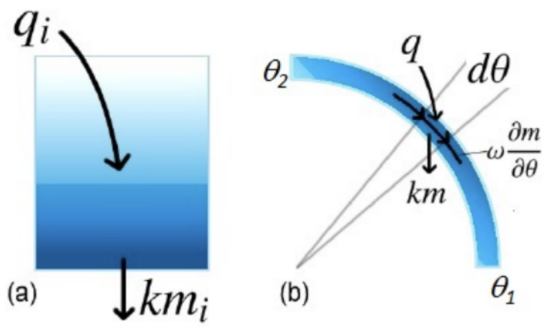


Figure 2: Continuous Water Wheel Model

The notion of a discrete wheel can expand to a continuous version. In this case, water in specific section $d\theta$ can move to other sections such that the frame of inference is fixed outside the rotation of the wheel.

Here, I am going to express the dynamics of a waterwheel mathematically and show how the movement of a waterwheel follows Lorenz Equation.

Before we start the section, let us define some notations.

$w(t)$ : angular velocity of the water wheel.
$\theta$ : angle in lab frame where $\theta = 0$ indicates 12:00 direction.
$Q(\theta)$ : The rate at which water is pumped in. (Not depend on $t$)
$r$ : radius of the wheel
$m(\theta, t)$ : mass distribution of water around rim.

## 1.2 Conservation Law

The mass of water between $\theta_1$ and $\theta_2$ is

$$M = \int_{\theta_1}^{\theta_2} m(\theta, t) d\theta \tag{1}$$

The conservation law of the mass of water in the wheel can be written as

$$\frac{\partial m(\theta, t)}{\partial t} = Q - Km(\theta, t) - w\frac{\partial m(\theta, t)}{\partial \theta} \tag{2}$$

where $Q$ is the amount of water in-flowing into the wheel and $Km$ is the amount of water leaking from the water wheel. $w\frac{\partial m(\theta,t)}{\partial t}$ is the amount of water transport from $\theta$ to the adjacent part. Here, I assume that the rate at which water leaking from the water wheel is constant.

Then, consider the sector where $\theta \in [\theta_1, \theta_2]$.

The mass between this section, $M$ can be expressed as

$$\Delta M = \Delta t \int_{\theta_1}^{\theta_2} \frac{\partial m(\theta, t)}{\partial t} d\theta = \Delta t \Big[ \int_{\theta_1}^{\theta_2} Q(\theta) d\theta - \int_{\theta_1}^{\theta_2} Km(\theta) d\theta - m(\theta_2)w + m(\theta_1)w \Big] \tag{3}$$

Physical interpretation is as follows.

$\Delta t \int_{\theta_1}^{\theta_2} Q(\theta) d\theta$ is the amount of water inflowing to the section $[\theta_1, \theta_2]$.

$\Delta t \int_{\theta_1}^{\theta_2} Km(\theta) d\theta$ is the amount of water leaking from the section.

$m(\theta_2)w\Delta t$ is the amount of water carried out to the section and $m(\theta_1)w\Delta t$ is carried into the section.

If the section is $[0, 2\pi]$,

$$\frac{\partial M}{\partial t} = Q_{tot} - KM \tag{4}$$

where $M$ is the total mass of water on the wheel and $Q_tot$ is the total rate at which water is pumped into the whole wheel.

## 1.3 Newton's law

Suppose $I(t)$ is the moment of inertia of the wheel. Then, the torque can be expressed as

$$\frac{\partial(Iw)}{\partial t} \tag{5}$$

There are two source terms that produce torque. One is the damping term, and the other is the torque due to gravity. If we assume that the damping torque is linear to the angular velocity, we have

$$\frac{\partial(Iw)}{\partial t} = -vw + gr\int_0^{2\pi} m(\theta,t)sin\theta d\theta \tag{6}$$

,where $v$ is the damping rate and $g$ is effective gravity.

The moment of inertia $I$ can be treated as a constant provided that enough time has elapsed and reached the stable phase. Therefore,

$$I\frac{dw}{dt} = -vw + gr\int_0^{2\pi} m(\theta,t)sin\theta d\theta \tag{7}$$

### 1.4 Fourier Series

Since $m(\theta,t)$ is $2\pi$ periodic in $\theta$, it can be represented as a Fourier time series.

$$m(\theta,t) = \sum_{n=0}^{\infty} a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta) \tag{8}$$

Similarly, $Q(\theta)$ also can be represented as Fourier series.

$$Q(\theta) = \sum_{n=0}^{\infty} q_n\cos(n\theta) \tag{9}$$

The reason why there is no sine term in $Q$ is because water falls into the wheel symmetrically at the top.

Now, we can plug Eqs.(8) and (10) into Eq.(2).

$$\begin{aligned}\frac{\partial}{\partial t}\sum_{n=0}^{\infty} a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta) = \sum_{n=0}^{\infty} q_n\cos(n\theta) - K\Big[\sum_{n=0}^{\infty} a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta)\Big] \\ - w\frac{\partial}{\partial\theta}\sum_{n=0}^{\infty} a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta)\end{aligned} \tag{10}$$

Since sine and cosine functions have orthogonality, we can calculate the coefficients respectively,

$$\dot{a_n} = -Ka_n + wb_n \tag{11}$$

$$\dot{b_n} = q_n - Kb_n - wa_n \tag{12}$$

Finally, we can plug Eq.(8) into Eq.(7).

$$\begin{aligned}\frac{\partial(Iw)}{\partial t} &= -vw + gr\int_0^{2\pi}[\sum_{n=0}^{\infty} a_n(t)\sin(n\theta) + b_n(t)\cos(n\theta)]\sin\theta d\theta \\ &= -vw + gr\int_0^{2\pi}[a_1(t)\sin(\theta)]\sin\theta d\theta \\ &= -vw + gra_1\pi\end{aligned} \tag{13}$$

Here, all the terms except $\sin\theta$ in $m(\theta, t)$ vanish because of the orthogonality.

In summary we get three equations, where $n$ is a non negative integer.

$$\boxed{\begin{aligned}\dot{a_n} &= -Ka_n + wb_n \\ \dot{b_n} &= q_n - Kb_n - wa_n \\ I\dot{w} &= -vw + gra_1\pi\end{aligned}} \tag{14}$$

In the above equation, only $a_1$ ,$b_1$, and $w$ are decoupled. Therefore,

$$\boxed{\begin{aligned}\dot{a_1} &= -Ka_1 + wb_1 \\ \dot{b_1} &= q_1 - Kb_1 - wa_1 \\ I\dot{w} &= -vw + gra_1\pi\end{aligned}} \tag{15}$$

This is exactly the same as the Lorenz equation after introducing the following equations and change of the time coordinate from $t$ to $s = Kt$

$$\rho = \frac{\pi A g r^2 q_1}{K^2 v} \tag{16}$$

$$\sigma = \frac{v}{KI} \tag{17}$$

$$\beta = 1 \tag{18}$$

and

$$x = \frac{w}{K} \tag{19}$$

$$y = \frac{\rho K a_1}{q_1} \tag{20}$$

$$z = \rho - \frac{\rho K b_1}{q_1} \tag{21}$$

### 1.5 Lorenz Equation

$$\boxed{\begin{aligned} \dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\sigma - z) - y \\ \dot{z} &= xy - \beta z \end{aligned}} \tag{22}$$

1) Lorenz equations have a symmetric pair of solutions $(x, y, z)$ and $(-x, -y, z)$

2) z-axis is invariant. The point on the z axis where $x = 0, y = 0$ remains on it.

3) Stationary Point. (I used $r$ instead of $\rho$ here) (Fig. 3)

• For $0 < r < 1$, Lorenz equation has only one stable fixed point.

• At $r = 1$, there is a bifurcation of a simple kind

• For $r > 1$, the origin is non-stable. There are two other stationary points which we shall call $C_1$ and $C_2$.

- For $\frac{(\sigma+\beta+3)\sigma}{\sigma-b-1} > r > 1$, $C_1$ and $C_2$ are stable.

- For $\frac{(\sigma+\beta+3)\sigma}{\sigma-b-1} < r$ , $C_1$ and $C_2$ are non-stable.

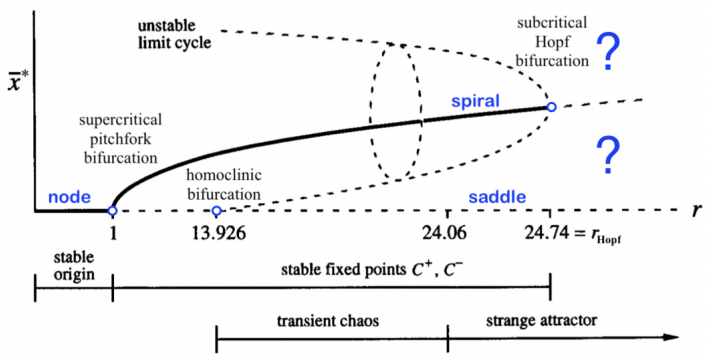


Figure 3: $\sigma = 10$ and $\beta = \frac{8}{3}$

# 2 Numerical Analysis

## 2.1 Euler method

The Euler method is a first-order numerical method for solving the initial-value problems of differential equations. It can be driven by Talyor's Theorem. Suppose that an initial value problem is given by

$$\frac{dy}{dt} = f(t, y) \quad \text{and} \quad y(t_0) = y_0 \tag{23}$$

Then

$$y(t+h) = y(t) + hf(t, y(t)) + \frac{h^2}{2}y''(\xi) \tag{24}$$

where $\xi$ is some number in $(t, t+h)$.

The Euler's method constructs $w_i \approx y(t_i)$, where $t_i = t_0 + i \cdot h$. Therefore, Euler's method is

$$\begin{aligned} w_0 &= \alpha. \\ w_{i+1} &= w_i + hf(t_i, w_i) \end{aligned} \tag{25}$$

Since this method is first-order method, the local error is $O(h^2)$ and the global error is $O(h)$.

## 2.2 Runge-Kutta method

Runge-Kutta method is a numerical method for solving the initial-value problems of differential equations.

Suppose that an initial value problem is given by

$$\frac{dy}{dt} = f(t, y) \quad \text{and} \quad y(t_0) = y_0 \tag{26}$$

Then the general Runge-Kutta method of order $m$ is

$$
\begin{aligned}
w_0 &= y_0. \\
w_{n+1} &= w_n + \sum_{i=1}^{m} c_i k_i. \\
k_i &= hf(t_n + \alpha_i h, w_n + \sum_{j=1}^{i-1} \beta_{ij} k_j).
\end{aligned} \tag{27}
$$

where $c_i$'s are constants and $\alpha_1 = 0$. $\alpha_i$'s and $\beta_i$'s are determined by expanding $y_{n+1}$ and $y_n$ about $t_n$ and $y_n$ and expanding $k_i$ about $t_n$.

The most well-known Runge-Kutta method is Runge 4th order(RK4) method. When $\alpha_2 = \alpha_3$,

$$
\begin{aligned}
\alpha_2 = \alpha_3 = \frac{1}{2} \qquad \alpha_4 = 1 \\
c_1 = \frac{1}{6} \qquad c_2 + c_3 = \frac{2}{3} \qquad c_4 = \frac{1}{6} \\
\beta_{32} = \frac{1}{6c_3} \qquad \beta_{42} = 1 - 3c_3 \qquad \beta_{43} = 3c_3
\end{aligned} \tag{28}
$$

We can choose $w_2 = w_3 = \frac{1}{6}$ and this makes the solution

$$
\begin{aligned}
w_0 &= y_0. \\
k_1 &= hf(t_n, w_n). \\
k_2 &= hf(t_n + \frac{h}{2}, w_n + \frac{1}{2}k_1). \\
k_3 &= hf(t_n + \frac{h}{2}, w_n + \frac{1}{2}k_2). \\
k_4 &= hf(t_n, w_n + k_3). \\
w_{n+1} &= w_n + \frac{1}{6}(k_1 + k_2 + k_3 + k_4)
\end{aligned} \tag{29}
$$

Since this method is a fourth-order method, the local error is $O(h^5)$ and the global error is $O(h^4)$.

# 3 Term Project Plan

## 3.1 Goal

1) Solve Lorenz system with Euler and RK4. Check the sensitive dependence on the initial condition where $r > r_{Hpof}$. Check whether we can be convinced with the numerical solution in a nonlinear system.

2) Solve Eq.(15) using 4th order Runge-Kutta method and check whether the Malkus-Lorenz

water wheel follows the Lorenz system.

## 3.2 Result

### 3.2.1 Lorenz System

Before directly calculating Eq.(15), I first solved Lorenz equation.

#### $RK4$

The numerical solution of the Lorenz system with RK4 is plotted in Fig. 4. Two Lorenz attractors appear in the graph because $r > r_{Hopf}$. The $r_{Hopf}$ in Fig. 4 is 15.

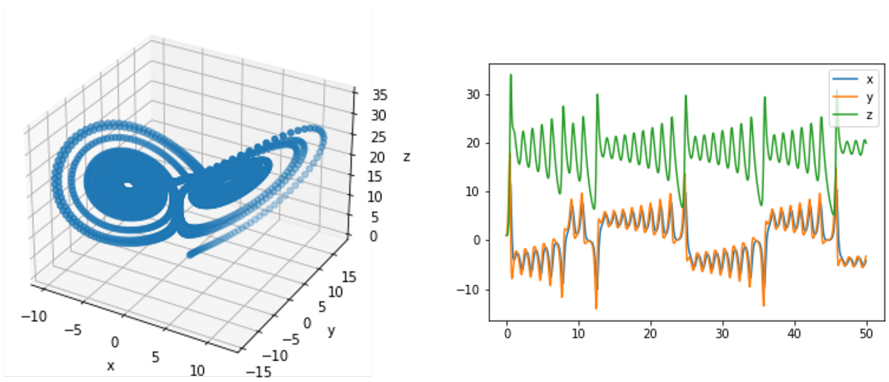


Figure 4: Solution with RK4. $\sigma = 5, \rho = 20, \beta = 1$.

I slightly changed the initial condition from $\rho = 20$ to $\rho = 20.00001$. The result is shown in Fig. 5. After $t$ becomes 35, the difference between two x values becomes very large.

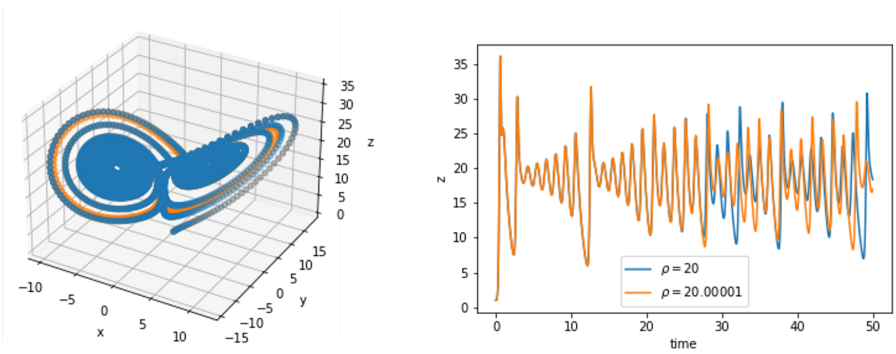


Figure 5: Initial condition sensitivity. *Orange*: $\sigma = 5, \rho = 20, \beta = 1$. *Blue*: $\sigma = 5, \rho = 20 + 0.00001, \beta = 1$.

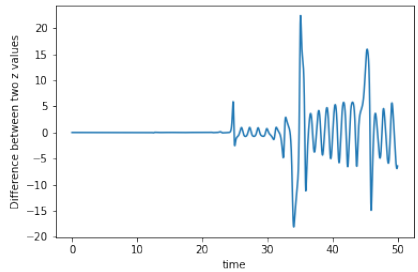


Figure 6: The difference between z values with same $\sigma$ amd $\beta$ but different $\rho$: $\rho = 20$ and $\rho = 20.00001$

### *Euler*

The numerical solution with the Euler's method is shown in Fig. 7. Two Lorenz attractors appear with appropriate parameters.

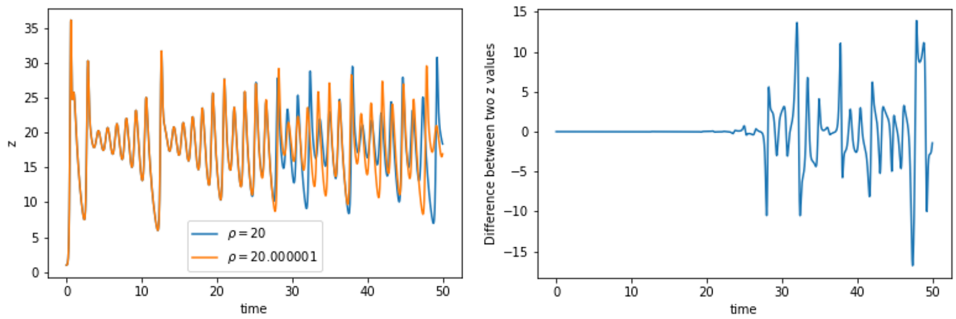


Figure 7: Initial condition sensitivity with Euler method. *Left*: numerical solutions with slightly different parameters. *Right*: Difference between two solutions. The difference diverge after about $t = 25$

### *Timestep sensitivity*

It is well known that numerical solutions are affected by the scale of a time step. I calculated the solutions with three different time steps with RK4, and the result is shown in Fig. 8.

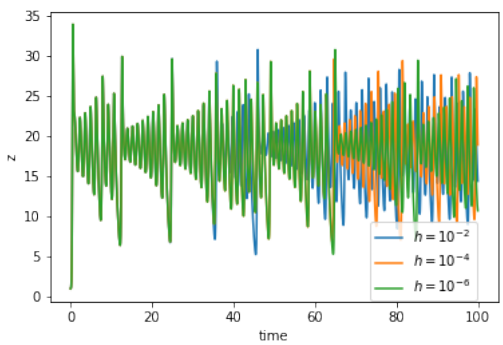


Figure 8: Numerical solutions of RK4 with three different time steps.

After $t$ becomes about 35, numerical solution for a time step $10^{-2}$ diverges from other solutions. All solutions diverge after $t$ is about 60. As shown in Fig. 9, the two solutions look completely

different after some extent of time passes.

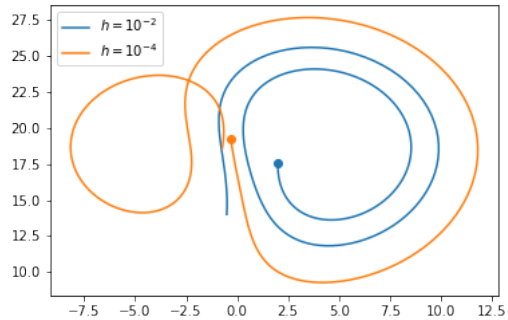


Figure 9: Numerical solutions for two different time steps with the same initial points become completely different in $95 \leq t \leq 100$. Two dots represent where $t = 95$.

I also repeated the same procedure with Euler's method. As shown in Fig. 10, the solutions diverge much earlier than the solutions with RK4.

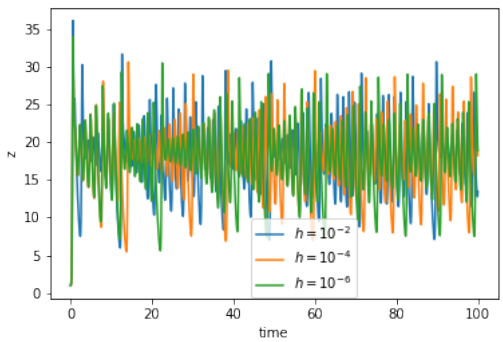


Figure 10: Numerical solutions of Euler's method with three different time steps.

### 3.2.2 Malkus-Lorenz waterwheel

#### *RK*4

The numerical solution of Malkus-Lorenz waterwheel with RK4 is shown in Fig. 11. The graph is sparser than the solution of the Lorenz system. This is because time coordinate $s$ in Lorenz system is $K$ times $t$, where $t$ is time coordinate in Malkus-Lorenz wheel system and $K < 1$.

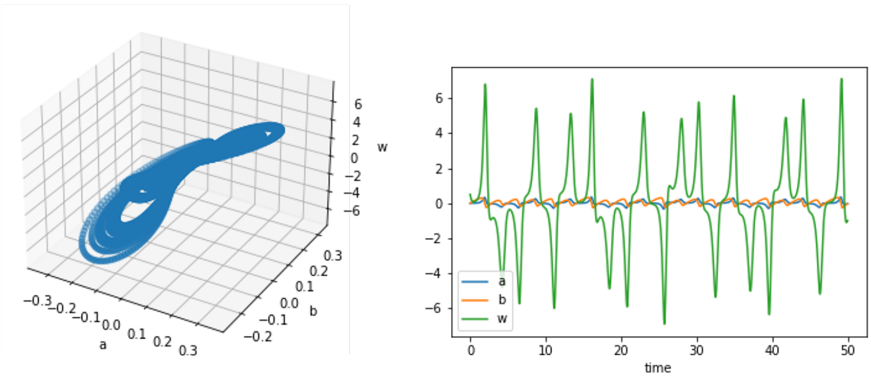


Figure 11: Numerical solution of Malkus-Lorenz water wheel with RK4 method. $v = 0.25, K = 0.5, r = 0.25$ and $I = r^2$. Initial point starts at $(0, 0, 0.5)$ and time step is $h = 0.01$.

### *Bifurcation*

Numerical solution is affected by time step scale. This also can be observed in bifurcation diagram.

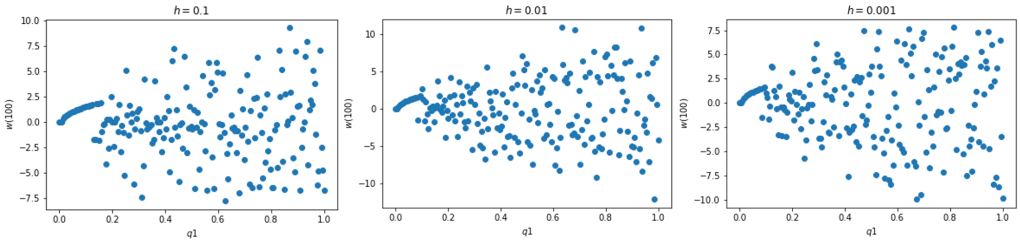


Figure 12: Bifurcation diagram with various time steps. RK4 method is used. Parameters and initial starting point are same with Fig. 11.

I zoomed in on the Fig. 12, to see how the bifurcation point changes when the time step changes. The bifurcation point is about 0.125 when $h = 0.1$, while the points are smaller when $h = 0.01$ or $h = 0.001$. Therefore, when $q_1$ is in $(0.1, 0.125)$, chaotic phenomenon appears with the time step is 0.01 and 0.001, but not with the time step is 0.1

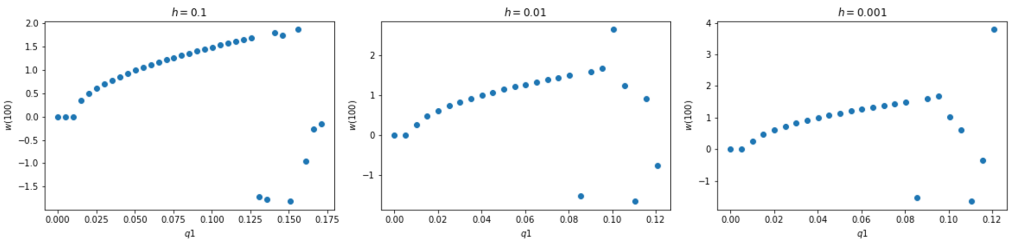


Figure 13: Bifurcation points with different time step scales.

### 3.2.3 Code Availability

You can check the code in https://github.com/MunjungKim/Malkus-wheel